

Differential Capacitance of Ionic Liquid Confined between Metallic Interfaces

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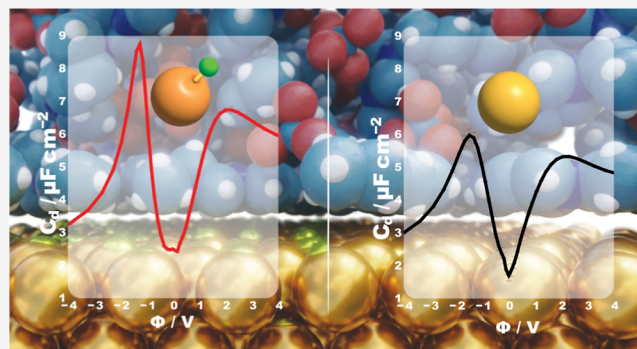


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ABSTRACT: We present here a detailed analysis of the electric double layer at the gold electrode/[BMIM][BF₄] interface using a polarizable model for the electrode, based on our recent approach to include image charges [Geada et al. *Nat. Commun.* 2018, 9, 716]. A double bell (camel) shape is obtained for the differential capacitance, where the inclusion of metal polarization allows for a higher density of ions in the double layer, particularly around the maxima, thereby increasing the capacitance. The charging mechanism differs for the positive and negative electrodes, with counterion adsorption prevailing at the anode and co-ion desorption prevailing at the cathode. The charging mechanism is predominantly governed by the BF₄ anions, serving as counterions and co-ions at the anode and cathode, respectively. Within the considered range of potentials, only minor changes are observed in the dynamical properties, specifically in the diffusion coefficients. Notably, it is interesting to observe that bulk properties are restored at a shorter distance from the gold surface in the case of the anode compared to the cathode.



INTRODUCTION

The last two decades have seen a strong development of research in ionic liquids (IL) due to their application as the electrolyte in electrochemical capacitors, or supercapacitors.^{1–4} A crucial step to advance such applications is the understanding of the structure of the electric double layer (EDL) at the electrolyte/electrode interface.^{5,6} Experimentally, information on the EDL structure can be indirectly obtained by measuring the differential capacitance,⁷ namely, the capacitance as a function of the applied voltage. A few features of the differential capacitance of ILs are well established, e.g., its asymmetry as consequence of the asymmetry in the constituent ions. Mean-field theory,^{8–11} classical density functional theory,^{12,13} as well as molecular dynamics (MD) simulations,^{9,14–17} and experiments¹⁸ have shown that the capacitance presents a higher value on the electrode containing the smaller ions. Another important aspect influencing the shape of the differential capacitance is the nonspherical shape of the ions, as well as the presence of charged heads and neutral tails. Both elements have been shown to be responsible for the camel-shape (or double-bell) of the capacitance as a function of the applied voltage. Indeed, in both cases, a nonspherical shape¹⁶ as well as the possibility to replace neutral tails with charged groups via rotation and translation¹⁵ provide extra degrees of freedom for the field-induced charge rearrangements in the double layer. The specific ion adsorption modifies the position of the potential of zero charge (PZC).

In the calculation of the differential capacitance two approaches are typically used, namely the constant charge,^{19,20} where a constant charge is uniformly distributed on the electrodes, and the constant potential approaches,^{21–26} where instead a constant potential is imposed on the electrode. In the fixed potential approach, the partial charges carried by the electrode atoms are treated as additional degrees of freedom, which can fluctuate during the simulation, and whose values are determined self-consistently for each electrolyte configuration. In the case of flat electrodes (surfaces), previous calculations on coarse-grained ILs confined between graphite electrodes²⁷ have shown that the two approaches provide similar, although not identical, results. Different is the situation of nanostructured electrodes, as e.g., carbon electrodes in supercapacitors where instead the presence of corrugated surfaces, channels, etc. can lead to strong charge localization and strong electric field heterogeneities.^{28–30}

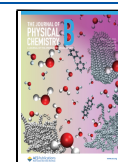
Recently we have introduced a simple model to include the metal polarization, which consists of a Lennard-Jones potential and a harmonically coupled core–shell charge pair for every

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metal atom. The model reproduces the classical image potential of adsorbed ions as well as surface, bulk, and aqueous interfacial properties in agreement with experiments.³¹ We have also extended the model to describe variable potential conditions beyond the potential of zero-charge conditions. This is obtained by combining the polarizable model with additional charges on the electrode surface. Our approach has been used to investigate the Au(111) surface in contact with a NaCl electrolyte solution within a wide range of interfacial electrostatic potentials, where we found that the metal polarization enhances the interfacial capacitance influencing the ions distribution at the electrode/electrolyte interface.³² In this contribution we apply our polarizable model for the metal in combination with additional fixed charges placed on surface atoms of the metal to simulate the ideal metallic behavior. We vary the potential via the surface charge density to calculate the differential capacitance for EDL at the Au(111) electrode/[BMIM][BF₄] interface. The motivation for such a choice is that, although such a system has been extensively investigated both on the experimental and on the computational point of view, some open questions still remain regarding the structure and dynamics of the EDL. When charging the electrode, the advantage of our model with respect to the traditional fixed charge model, is that it allows to include local polarization effects, as for example, those induced by the presence of local charge heterogeneity. Charges of opposite sign are evenly distributed on the surface of two electrodes, which confine the IL. During the MD simulation, for each configuration, every surface gold atom carries a net charge and an instantaneous dipole moment. A question may arise if the extra $\pm\Delta q$ charge should sit on the core (gold atom) or on the dummy. We have considered both possibilities, and we do not observe significant differences between them. In the current manuscript, we will always refer to the charge-on-core case, while the detailed comparison between the two cases can be found in the [Supporting Information](#).

The paper is organized as follows: the structure of the double layer is investigated using atomistic simulations within our polarizable model for the metal surface³¹ and the results compared to the fixed charge approach. In particular, the details of the differential capacitance are discussed for both cases. The charging mechanism, as well as the structural evolution, as a function of the interfacial electric potential, will also be presented. Along with structural properties, dynamical properties will also be discussed, particularly, the self-diffusion coefficient.

METHODOLOGY

Our model system contains 1200 pairs of [BMIM][BF₄], confined between two gold electrodes each made of 9 layers of gold atoms in the (111) crystallographic plane and spanning the 4.055×4.055 nm² cross-section of the box. Outside the electrodes, a vacuum region of 222 nm is added. Two-dimensional periodic boundary conditions are enforced along the x and y directions, while the interaction between images along the z -directions are decoupled using the Particle Mesh Ewald with slab correction.³³ To describe the atomistic IL, we use a nonpolarizable model of [BMIM][BF₄] with force field parameters that reproduce experimental heat of vaporization, diffusion coefficient, ionic conductivity, and shear viscosity of [BMIM][BF₄].³⁴ An extensive characterization of such a system at the PZC has been presented in our previous work.³⁵

Two models for the gold were considered, a nonpolarizable gold model³⁶ and a polarizable one,³¹ which includes the induced image charges. In order to obtain variable potentials for the electrodes, charges of opposite signs are evenly distributed on the two surfaces that confine the IL; $+\Delta q$ on the anode and $-\Delta q$ on the cathode, with the result of an overall neutral system. As a net result, for each configuration, every topmost-layer gold site carries a net charge and an instantaneous dipole moment. The net charge at each site of the topmost layer is obtained by an equal distribution of total surface charge that yields a desired surface charge density. We performed, for each gold model, 23 simulations at different charge densities. The charge densities are obtained from $\sigma = \sinh((0.05x)^2)e \text{ nm}^{-2}$; $x \in [1, 23]$ to give values ranging from 0.0025 to 1.7432 $e \text{ nm}^{-2}$ and to allow for a higher density of points near $\sigma = 0 \text{ e nm}^{-2}$. Charge interactions in the gold are limited to those between different sites, excluding on-site interactions between bonded Au-core and Au-dummy.

To address the difference in the dynamical properties of our systems, we calculate the 2D diffusion coefficient in the plane of the gold surfaces within bins between gold surfaces. The calculation of the diffusion coefficient is performed with the same approach as in our previous work³⁵ such that given a bin, α , the 2D diffusion coefficient is calculated according to the Einstein relation

$$D_\alpha = \frac{1}{4} \lim_{\tau \rightarrow \infty} \frac{d}{d\tau} \text{MSD}_\alpha(\tau) \quad (1)$$

where MSD_α is the mean squared displacement within the bin α , defined, for example along the x -axis and with a reference time t , as

$$\text{MSD}_{x,\alpha}(t, \tau) = \frac{\sum_i \sigma_{i,\alpha}(t) \sigma_{i,\alpha}(t + \tau) (x_i(t + \tau) - x_i(t))^2}{\sum_i \sigma_{i,\alpha}(t) \sigma_{i,\alpha}(t + \tau)} \quad (2)$$

where $\sigma_{i,\alpha}(t)$ is a switching function for the i th particle in the bin α at time t , and $x_i(t)$ is the coordinate of that i th particle at time t . The switching function allows to account for particles entering and exiting a bin during the course of the simulation. Then by restarting the calculation of $\text{MSD}_{x,\alpha}(t, \tau)$ at different reference times

$$\text{MSD}_\alpha(\tau) = \frac{1}{N} \sum_{n=1}^N \sum_{j \in \{x,y\}} \text{MSD}_{j,\alpha}(n\Delta t, \tau) \quad (3)$$

where Δt is a restart time, which is the time between two successive reference times, and N , is the number of restarts.

In order to ensure a statistically relevant sampling, five independent initial configurations were sampled from an equilibration run of 20 ns at a constant temperature of 298 K using a Nosé–Hoover thermostat, and at a constant pressure of 1 bar using a semi-isotropic Berendsen barostat. Therefore, for each value of the charge on the electrode, 5 independent realizations are considered.

All simulations were performed with version 5.1.4 of the GROMACS simulation package.³⁷

RESULTS AND DISCUSSION

At the point of zero charge (PZC), when no net charge is present on the surface, an overscreening effect is observed, due to the size and shape difference between cations and anions. In particular, the cations adsorb closer to the gold surface than

the anions, with the imidazole ring parallel to the gold surface. For the employed model (including the metal polarization) this leads to a PZC, $\Phi_0 = 0.26$ V. As discussed in our previous work,³⁵ the inclusion of the image charges for the metal does not have a big impact on Φ_0 .

It is interesting to compare the differential capacitance, C_d , for the nonpolarizable and the polarizable gold models. C_d is defined as

$$C_d = \frac{\partial \sigma(\Phi)}{\partial \Phi} \quad (4)$$

where $\Phi = \Phi_{\text{electrode}} - \Phi_{\text{bulk}} - \Phi_0$, where $\Phi_{\text{electrode}}$ and Φ_{bulk} are electrostatic potentials at the surface of the electrode and in the liquid bulk, respectively. The electrode surface for polarizable gold was determined from the density profiles as the edge of the average sphere, centered on the topmost Au-core atoms, within which lies the topmost Au-dummies. In Figure 1, we

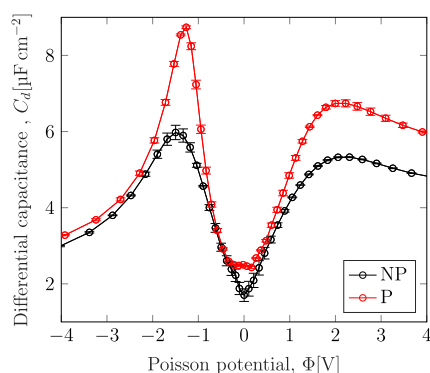


Figure 1. Differential capacitance for the atomistic simulations of the gold/[BMIM][BF₄] interface with the nonpolarizable gold model (black) and the polarizable gold model (red). According to the definition in eq 4, the potentials are referred to their corresponding values at PZC.

report the differential capacitance curves calculated for the polarizable (red) and nonpolarizable (black) models. Both curves present a minimum at 0 V (which, according to our definition, corresponds to the PZC) and increases for more positive or more negative values of the potential thus resulting into the double bell shape. Such a shape has been experimentally observed in imidazole-based $n = 2, 4, 8$ BF₄ ILs at the interface with platinum and polycrystalline gold electrodes,^{9,18,38} as well as at glassy carbon.³⁹ Interestingly the values for C_d are very close to the experimental data. Only the potential range in our case appears to be stretched to larger values with respect to the experiments.³⁸ The double bell shape has also been predicted with models.^{8,9,40–42}

The largest differences between the polarizable and nonpolarizable models are observed at moderate positive and negative potential values, in the area around the maxima. In particular, the polarizable model provides higher values for the capacitance, suggesting that for a given value of the electrode potential, the system is capable of storing more charge. In other words, the addition of the metal polarization permits to the system to store more charge in the double layer. It is also interesting to notice that at negative values of the potential there is an important difference in the capacitance between polarizable and nonpolarizable models around the maximum. For more negative values of the potential (below -2 V) however, the results of polarizable and nonpolarizable force

fields agree. On the other hand, the situation is different for the positive potential branch. Indeed the difference between polarizable and nonpolarizable models, which develops around the maximum, also persists for more positive values of the potential (above 2 V).

The shape of the differential capacitance is not symmetric for positive and negative values of the potential. In particular, a higher capacitance is found for negative potentials. A similar behavior was also observed in the experiments³⁹ and simulations⁴³ of for [BMIM][PF₆] on graphite, where cations are the same as those investigated here and the electrode surface is flat, similarly to the (111) gold surface. The asymmetry between the two branches of the capacitance is related to the difference in the ions size and shapes and to the different mechanisms, which is observed for the double layer formation that we will discuss in detail here below.

When the electrode is charging, the charge can be balanced by two processes, namely the adsorption of counterions (ions with opposite charge with respect to that on the electrode) and the desorption of co-ions (ions with the same charge sign as that of the electrode). The competition between these two processes determines the local density at the interface in the process of charging. To describe the charging mechanism we use the same parameter X introduced by Merlet and co-workers in the case of the porous carbon electrodes,⁴⁴ with the difference that here we use it to characterize the first adsorbed layer on the electrode. The parameter X is defined as

$$X(\Phi, \Phi_0) = \frac{N(\Phi) - N(\Phi_0)}{(N_{\text{counter}}(\Phi) - N_{\text{co}}(\Phi) - (N_{\text{counter}}(\Phi_0) - N_{\text{co}}(\Phi_0)))} \quad (5)$$

where $N(\Phi)$ is the total number of ions in the first adsorbed layer at a given charging potential Φ , $N(\Phi_0)$ is the total number of ions in the first adsorbed layer at PZC, Φ_0 . $N_{\text{counter}}(\Phi)$ and $N_{\text{co}}(\Phi)$ are the number of counterions and co-ions, respectively, at the given potential Φ . The calculation of X parameters was performed in the region within 3.8 Å of the surface of the gold slabs. Such a cutoff was chosen to only include the first ions' layer at the interface, which is the one that is most affected by the polarization. X provides a measure of the relative balance between counterion adsorption and co-ion desorption in the charging mechanism. When the counterion adsorption prevails, X is positive, for ion exchange, $X = 0$, and finally, if the co-ion desorption prevails, X is negative. Such a definition also implies that X is confined to values between -1 and $+1$. In our case we find that at the anode (positive potential) X is close to $+0.5$, suggesting that counterion adsorption prevails during charging. On the other hand, at the cathode (negative potential), X is close to -0.5 suggesting that co-ions desorption prevails (Figure 2). It is worth noting that BF₄ (the anions) are the counterions and co-ions at the anode and the cathode, respectively, and therefore in the case of [BMIM][BF₄] the charging mechanism is largely driven by the anions.

Also, in Figure 2 we report how X varies as a function of absolute Poisson potential, $|\Phi|$ and we compare the results for the traditional (nonpolarizable) fixed charge approach and for our polarizable electrode. We find that the absolute values of X are smaller for the polarizable model (red lines), suggesting a larger weight of the ion exchange over the adsorption/desorption of ions in the case of the polarizable model. In

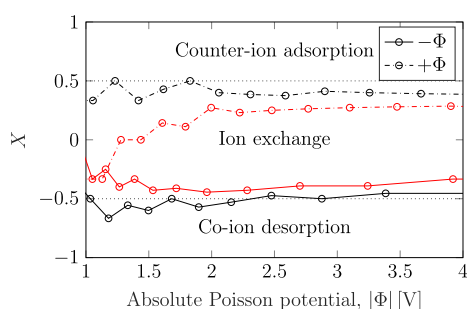


Figure 2. X parameter calculated within 3.8 Å from gold surface as a function of the interface potential. Black and red lines are for nonpolarizable and polarizable gold models, respectively, while solid and dashed lines are for negative and positive potentials (or surface charge densities). Horizontal dotted lines indicate boundaries separating the various charging mechanisms.

other words, especially around the maxima in the differential capacitance curves, ion exchange is enhanced in our polarizable gold. This well explains the enhancement seen in the differential capacitance curves since the ion exchange at the interface would result in higher ratios of counterions over co-ions stored at the interface (see Figure S7, showing how the ion ratios within 3.8 Å evolve with potential).

As the cations have a nonspherical symmetry, it is also interesting to investigate how their orientation changes as a function of the different charging states of the electrodes. In Figure 3 we show the distributions of the angles made by the relevant molecular axes on the cations and the normal to the electrodes surface, namely θ , the angle made by the axis across the imidazole ring, ϕ , the angle made by the axis along the butyl tail, and ψ , the angle made by the plane of the imidazole ring.

For the positive electrode, with increasing Φ the cations lose the mostly flat orientation which is observed at the PZC and are more disordered. This is shown, for example, by the angle distributions in Figure 3, where the peaks in panels A2, B2, and C2 are gradually lost as the potential increases.

As the cations move from the flat geometry on the surface, they leave space in exchange for the counterions, which indeed is suggested by the X parameter. On the other hand, on the negative electrode, the axis across the imidazole ring (panel A1 in Figure 3) remains parallel to the electrode surface. In addition to this, the stable peaks in panels B1 and C1 indicate a stable flat orientation for the cation (BMIM) especially at

lower potentials. Similarly, sum frequency generation (SFG) spectra had suggested for [BMIM][DCA] that the imidazolium ring in the cation tends to lie flat at the surface of the electrode at negative potentials but is repelled at positive ones.⁴⁵ Flat-lying cations at the negative potential maximize the attractive interaction between the cations and the electrode, whereas, on the other hand, at the positive potential, cations tilt along the surface normal as they are repelled to maximize attractive interaction between anions and electrodes.⁴⁶ Similar reorientation of cations at the anode upon increasing electrode potential has also been shown by Motobayashi et al.⁴⁷ using surface-enhanced infrared absorption spectroscopy.

While the reorientation at the anode is very well established, it is interesting to note that there is also a reorientation of cations at the cathode upon increasing electrode potential. The stability of peaks in panels B1 and C1 of Figure 3 is not as maintained throughout the charging of surface as it is strongly maintained in panel A1. At the higher potentials, the tail orientation becomes randomized but within a range (panel B1). At the same time, the head shows, by the appearance of a plateau centered on $\pi/2$ (panel C1), a more upright configuration. At such high potentials, this picture of the cations mostly lying flat at the cathodic interface begins to change. With increasing Φ , the probability of an imidazolium cation having either a raised butyl tail or a tilted imidazole plane also increases (panel B1 and C1 in Figure 3). Kemna and Braunschweig⁴⁸ using in situ SFG experiments showed a more upright orientation of the imidazolium ring (of EMIM in their case) at a Pt/[EMIM][BF₄] interface (in the presence of 0.5 M H₂O) for both negative and positive potentials with respect to the PZC, for which the orientation was rather more flat. Ratschmeier and Braunschweig⁴⁹ further found that at very high negative potentials, the cation ([BMMIM]) showed a more upright orientation. This configuration allowed for creation of voids that were then occupied by water molecules present in their system. While comparing their results to previous studies under dry vacuum and in the absence of water, Ratschmeier and Braunschweig⁴⁹ showed that the presence of water could be an important determinant of the potential dependence of imidazolium orientation at an electrode interface. In our study, however, we find similar potential dependent orientation even in the absence of water. The molecular reorientation, in our case, also permits to increase in

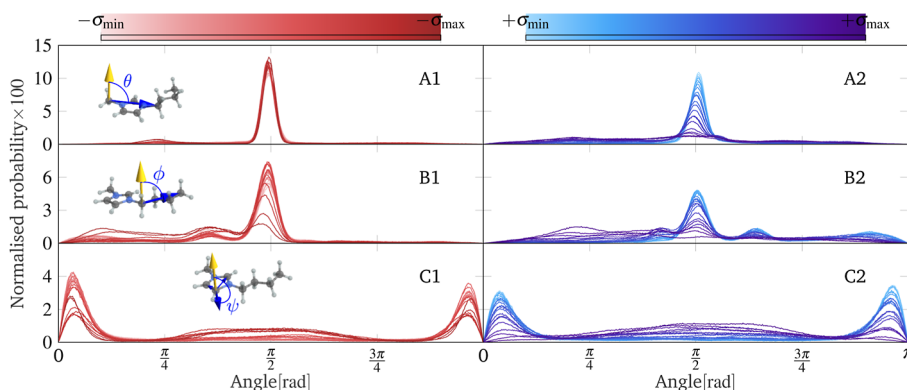


Figure 3. Orientation of the cations within 5 Å from polarizable gold surface. Panel labels (A–C) represent the distributions of the angles θ , ϕ , and ψ respectively. The left panel (red lines) and the right panel (blue lines) correspond to the negatively and positively charged surfaces, respectively.

the density of the adsorbed cations on the surface as the applied negative potential increases.

In Figure 4 we present the changes in the molecular orientation in the first adsorbed layer on the cathode, at

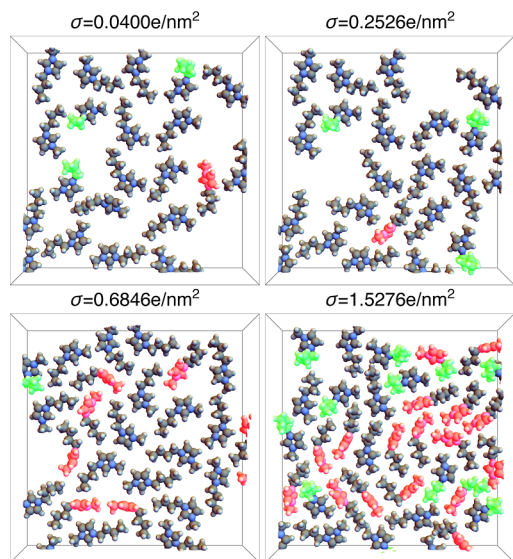


Figure 4. Orientation of cations at cathode. Upon charging, two main orientational adjustments to permit crowding are identified—raised tails, shown in green and perpendicularly tilted imidazole ring, shown in red.

different values of increasing surface charge (potential). In particular, the molecules whose butyl tails are colored in green show a change in the butyl tail orientation, which moves from laying on the surface to standing up ($\phi < \pi/4$) as the absolute electrode potential increases. The red-colored imidazolium rings indicates instead those molecules whose imidazole plane changes orientation from laying flat and parallel to the surface to standing orthogonal at the interface. Since the tail is neutral while the head/ring is charged, both types of change in the molecular orientation increase the ratio of imidazolium charge to the occupied interfacial area, allowing for a high counterion density in response to the very high negative potentials. It is well understood that the asymmetries in these ILs are the precursors for peculiar properties ILs exhibit. A direct consequence of the asymmetries in shape and size is the packing pattern that develops under certain conditions. The packing pattern that we find here is not only due to the asymmetries in shape but also to the flexibility of the ions. The consequences thereof throw a reasonable challenge to the modeling of ILs as charged spheres or rigid bodies especially at an interface, since this “softness” could not be captured in spherical and even nonspherical but highly coarse-grained rigid models.

Finally, we would like to briefly comment on the changes in the dynamical properties during the charging of the electrode. To this purpose, we have calculated the diffusion coefficient as a function of the distance from the gold surface. For each value of the charging potential, we calculate here the 2D translational self-diffusion in the plane of the gold surface within bins between the two gold surfaces, for the two ionic components. Figure 5 shows the diffusion coefficient across the interface for BMIM (blue tones) and for BF_4 (red tones). The values at the PZC are highlighted in black (dashed line for cations and solid

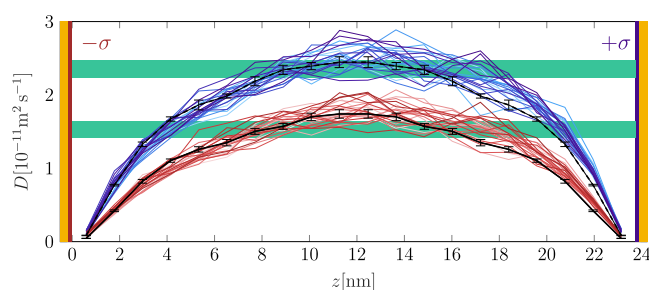


Figure 5. Diffusion coefficient of cations (in blue tones) and of anions (in red tones) across the interface. The black curves are for cations (dashed) and anions (solid) at PZC. Green strips are bulk diffusion coefficients of the ions with error bars.

line for the anions). The bulk diffusion coefficient of the ions, calculated from five independent trajectories of 800 ion pairs at 298 K and including the size correction⁵⁰ is also reported, including the statistical uncertainty as a green stripe. The cations diffuse much faster than the anions, with values monotonically increasing to the bulk value as we move away from the gold surface. We would like to stress that, as already pointed out in our previous work,³⁵ that although from the structural point of view, the EDL is only a couple of nm thick, from the dynamical point of view the perturbation to the diffusion coefficient persists on a much wider scale of the order of 10 nm. The main result here is that increasing the charge on the electrode does not significantly affect the diffusion coefficients, which remain very close to the values calculated and discussed at the PZC. Nonetheless, a slight asymmetry can be observed between the two interfaces (see Figure 5). In particular, at the cathode/liquid interface (negative potential side) the ions recover more slowly (at larger distances from the surface) the bulk behavior, while at the anode/liquid interface (positive potential side) recovery of bulk properties is faster (within shorter distance from the surface when compared to the cathode). The diffusion being significantly slower at the interface than in bulk is in direct agreement with works from several authors.^{51–54} The asymmetry, however, may be an interplay of several factors. For example, there is already an asymmetry in the diffusion between cations and anions—cations being the more diffusive. Thus, as more cations are repelled at the anode (owing to ion ratios determined by the charging mechanism), the average diffusion near the anode tends to increase. However, we do not observe significant shift in the asymmetry for increasing potentials because the cations are also more resilient to repulsion at the interface for lower potentials as shown in their density profiles in Figure S6 and for higher potentials, the charging mechanism (X parameter) is fairly constant.

CONCLUSIONS

We have presented here a detailed analysis of the EDL at the gold electrode/[BMIM][BF_4] interface. In particular, we have investigated the shape of the differential capacitance comparing a polarizable model for the gold electrode, based on our recent work^{31,32} and the standard fixed charge approach. The polarizable electrode provides higher values of the differential capacitance allowing to pack a higher density of ions in the double layer. Such a difference is particularly pronounced at the maxima of the double bell, and on the positive potential branch above 2 V. The charging mechanism is different for the positive and negative electrodes. In particular, at the anode

(positive potential) counterion adsorption prevails during charging. On the other hand, at the cathode (negative potential), co-ions desorption prevails. In both cases, the charging mechanism is governed mostly by the BF_4 anions, which are the counterions, respectively co-ions at the anode and the cathode. We have also investigated the ions dynamics as a function of the applied potential and we have found that, within the considered range of potentials, there is no substantial change in the diffusion coefficients as a function of the distance from the interface, although it is interesting to notice that bulk properties are recovered at shorter distance from the gold surface in the case of the anode. As a future step, it will be interesting to test our polarizable model for the electrode on more complex geometries, to understand to what extent it is able to account for heterogeneity in the local electric fields.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpcb.3c08042>.

Additional information on the gold model; comparison of differential capacitances for the charge on core versus charge on dummy gold setup; and counterpart figures for traditional gold model (PDF)

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Notes

The authors declare no competing financial interest.

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